Online Appendix to

"The Relative Wage and Marriage after 2010: A Comment on Shenhav (2021)"

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August 2026

This appendix documents the reconstruction, the evidence on the constructed relative wage, the robustness checks, the secondary outcomes, and the supporting exhibits. It is intended to be read with the comment. A complete reproduction of the original tables and figures is provided in the replication package rather than here, and Section OA.6 describes what that package contains.

OA.1 Replication Details and Sensitivity

Columns 1 and 2 of Table 1 in the comment report the central replication exercise. They place our estimates, rebuilt from raw IPUMS microdata for 1980 to 2010, next to the published coefficients in Table 2 of Shenhav (2021), for the specification that uses the relative wage alone and for the one that adds the average potential wage. Point estimates, standard errors, and significance levels all lie close to the published values. Without controls our marriage coefficient is -0.0428 with a standard error of 0.0076, against a published -0.0428. With the controls measured at the market level it is -0.0502 with a standard error of 0.0086, against a published -0.0513.

Sources of divergence. The remaining gaps are consistent with differences in data vintage and sample coverage. On cells that both files contain, median differences are negligible and correlations exceed 0.99. The current extract nevertheless contains 253 cells absent from the author's file and omits 1,815 cells that the author's file contains, and coverage of the historical CPS weights and of the 2000 Census bridge sample also differs from the original extraction. We do not assign a directional bias to these gaps. The headline estimates reproduce closely in any case, and Section OA.3 reports the sensitivity to the controls. An age restriction defined consistently across cohorts leaves the substantive conclusions unchanged.

OA.2 Evidence on the Constructed Measure

Section IV of the comment shows that the 1970 weighted potential wage came apart from the observed relative wage within marriage markets after 2010. This section reports the evidence behind that claim. The break is specific to the relative wage, and to the difference in differences variation that identifies the coefficient. It is not present in the two wage levels separately, and it is not the survey redesign. The estimates from the later waves therefore say nothing about how women respond, in either direction.

OA.2.1 The measurement check and the ladder

Table OA.1 reports the measurement check at each of the four levels of saturation used in Table 3 of the comment, together with the share of the constructed measure's variance that survives the projection onto the fixed effects. The ladder shows two things. First, the check for the relative wage is a well measured zero in the later waves at every rung, including the lightest, so the collapse is not produced by the saturation of equation (1). Second, the residual variance share is no smaller in the later waves than in the original period, at 1.2 percent against 0.9 percent under the full specification. The fixed effects remove almost everything in both windows. Over 1980 to 2010 the same check returns $+0.593$ under the full specification, and the measure tracks the wage women face.

A note on a check that does not do the work it appears to do. Across marriage markets, and in levels, the constructed female potential wage correlates with the observed female wage at $+0.85$ in the later waves, and a regression of one on the other returns a slope above unity. Neither feature survives contact with the estimating equation. Equation (1) absorbs permanent differences across markets through the fixed effects, and it uses the relative wage rather than the female wage. A check computed in levels, across markets, and on the female wage therefore speaks to none of the variation that identifies β. We report it so that a reader who computes it is not misled by it. Computed on the within-market relative wage variation that does identify β, the measure has come apart.

Table OA.1: The Measurement Check at Four Levels of Saturation

	Relative wage		Female wage		Residual variance share	
Specification	1980–2010	2015–2024	1980–2010	2015–2024	1980–2010	2015–2024
(1) Full equation (1)	0.593***	−0.029	0.629***	0.211*	0.9%	1.2%
	(0.191)	(0.142)	(0.160)	(0.121)		
(2) Drop state by year	0.619***	+0.016	0.201	0.153	1.1%	1.5%
	(0.176)	(0.126)	(0.252)	(0.100)		
(3) Also drop race by state trends	0.723***	+0.047	0.305	0.093	1.6%	1.8%
	(0.210)	(0.109)	(0.201)	(0.154)		
(4) Market and year fixed effects only	0.380***	−0.012	2.485***	0.137	6.5%	3.0%
	(0.104)	(0.075)	(0.263)	(0.084)		
Observations	1,064	1,326	1,064	1,327		
Marriage markets	266	266	266	266		

Each entry regresses the observed series for a marriage market on the constructed series at the indicated level of saturation. The unit is the marriage market by year, so the birth cohort by year fixed effects of equation (1) in the comment are inapplicable, as they are in the corresponding exercise in the original study. Row (1) otherwise reproduces equation (1). Row (2) removes the state by year fixed effects, row (3) also removes the race by state linear trends, and row (4) retains only marriage market and year fixed effects. The residual variance share is the fraction of the variance of the constructed relative wage that survives the projection onto the fixed effects in that row, and it is reported to show that the fixed effects do not absorb differentially more in the later window. Standard errors in parentheses are clustered by state and cells are weighted by female population. The check for the relative wage is the one that bears on the identification of β, because equation (1) uses the relative wage and removes every permanent difference across markets. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

OA.2.2 Estimates inside the original sample

The measurement evidence dates the loss of identifying variation to the closing of the gender wage gap rather than to the boundary of our extension. If that is right, the relationship should already be weaker at the end of the original sample. Table OA.2 estimates equation (1) on pairs of adjacent waves inside 1980 to 2010. The 1990 and 2000 pair returns a significant marriage coefficient of

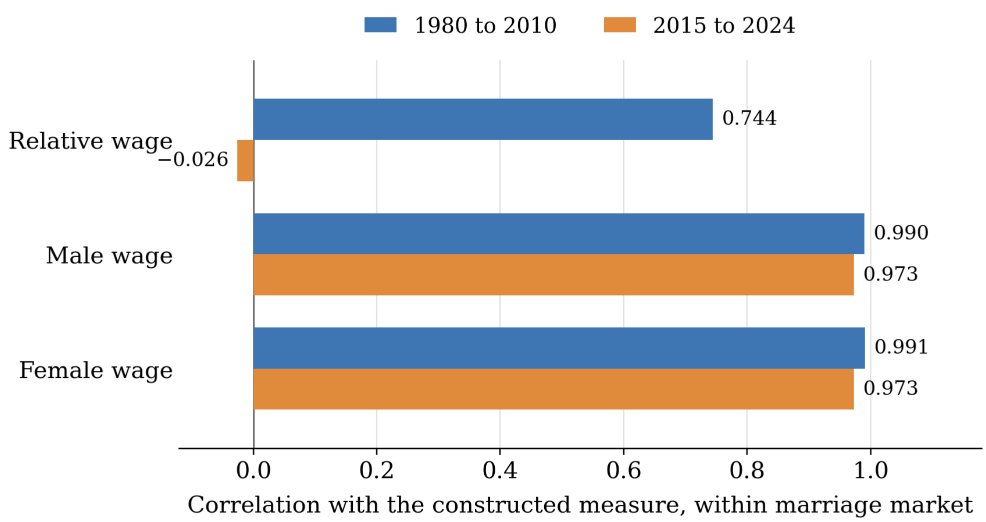

Figure OA.1: Both wage levels remain well measured after 2010. Their difference does not. Each series is deviated from its own marriage market mean within the window before the correlation is formed, and the correlation is weighted by female population.

−0.0616 with a standard error of 0.0172. In contrast, the 2000 and 2010 pair returns −0.0016 with a standard error of 0.0120, which is the same null we report for the later waves. Two waves are few and the standard errors are wide. The pattern nevertheless suggests that the attenuation began inside the original sample rather than at the boundary of our extension.

Table OA.2: The Relative Wage and Marriage on Pairs of Adjacent Waves inside the Original Sample

Waves used	Share married	Share divorced	Share never married
1980, 1990, 2000, and 2010	−0.0428***	+0.0231***	+0.0183***
	(0.0075)	(0.0049)	(0.0065)
1990 and 2000 only	−0.0616***	+0.0337**	+0.0239*
	(0.0172)	(0.0130)	(0.0136)
2000 and 2010 only	−0.0016	+0.0114	−0.0111
	(0.0120)	(0.0076)	(0.0098)

Each row estimates equation (1) of the comment on the waves indicated, using the specification without controls measured at the market level. Standard errors in parentheses are clustered by state and cells are weighted by female population. Coefficients are scaled to a 10 percent increase in the relative wage. The first row reproduces the full-sample estimate for the original period. The 2000 and 2010 pair falls entirely inside the sample of the original study. Two waves are few and these standard errors are wide, so we report the row as consistent with the measurement evidence rather than as an independent finding. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

OA.2.3 The observed relative wage as the regressor

Table OA.3 replaces the shift-share potential wage with the observed relative wage and re-estimates the three headline outcomes. The observed relative wage predicts none of them in either period. Over 1980 to 2010 the marriage coefficient is −0.0045 with a standard error of 0.0042. This needs care, and we build no part of the argument on it. The observed wage is endogenous, which is the entire reason for the shift-share construction, and selection into the labor market biases the coefficient toward zero. The table is therefore not evidence that no relationship exists. What it does show is narrower. The published finding rests wholly on the constructed measure and has no counterpart in the wages women are observed to face.

Table OA.3: The Observed Relative Wage Substituted for the Constructed One

Window	Regressor	Share married	Share divorced	Share never married
1980–2010	Constructed	−0.0428***	+0.0231***	+0.0183***
		(0.0075)	(0.0049)	(0.0065)
	Observed	−0.0045	+0.0024	+0.0022
		(0.0042)	(0.0023)	(0.0031)
2015–2024	Constructed	−0.0045	−0.0075*	+0.0116
		(0.0099)	(0.0041)	(0.0080)
	Observed	+0.0005	+0.0022*	−0.0027
		(0.0022)	(0.0012)	(0.0021)

Each entry estimates equation (1) of the comment with the regressor indicated, using the specification without controls measured at the market level. The constructed rows are the shift-share measure used throughout; the observed rows substitute the relative wage actually recorded for the marriage market in the same wave. Standard errors in parentheses are clustered by state and cells are weighted by female population. Coefficients are scaled to a 10 percent increase in the relative wage. The observed relative wage is endogenous, which is the reason the original study builds a shift-share measure in the first place, and simultaneity biases these coefficients toward zero. The table is therefore not evidence that no relationship exists. It establishes only that the published finding rests wholly on the constructed measure and has no counterpart in the wages women are observed to face, in either period. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

OA.2.4 Correlations of long differences

The coefficient in equation (1) is not identified from levels. It is identified from differential movement across marriage markets, so the relevant summary of the measure's quality is how well changes in the constructed series track changes in the observed series over comparable spans. Table OA.4 reports those correlations. In levels the fit across marriage markets decays gradually, from 0.61 over 1990 to 2010 down to 0.44 by 2024, which is the signature of a weighting scheme going out of date rather than of an abrupt break. In changes, however, the measure was never strong. The correlations over the decades that identify the published result run between 0.12 and

0.28, and over 2010 to 2024 the correlation is −0.021. Relevance in differential changes was low to begin with, and after 2010 there is none left.

Table OA.4: Correlation between Changes in the Observed and Changes in the Constructed Wage, across Marriage Markets

Change over	Span	Years since 1970	Relative wage	Female wage	Markets
Panel A: comparable spans					
1980 to 1990	10	15.0	+0.159	+0.489	266
1990 to 2000	10	25.0	+0.115	+0.136	266
2000 to 2010	10	35.0	+0.275	+0.185	266
2010 to 2019	9	44.5	−0.150	−0.192	264
2010 to 2024	14	47.0	−0.021	+0.087	266
2015 to 2024	9	49.5	+0.010	+0.078	265
Panel B: adjacent waves					
1980 to 1990	10	15.0	+0.159	+0.489	266
1990 to 2000	10	25.0	+0.115	+0.136	266
2000 to 2010	10	35.0	+0.275	+0.185	266
2010 to 2015	5	42.5	−0.021	−0.098	266
2015 to 2017	2	46.0	+0.188	+0.121	264
2017 to 2019	2	48.0	−0.146	−0.020	264
2019 to 2022	3	50.5	+0.086	+0.162	264
2022 to 2024	2	53.0	+0.110	+0.209	266

Each row correlates the change in the observed wage of a marriage market with the change in the constructed wage over the same interval, across marriage markets, weighted by female population. Years since 1970 is the age of the base year at the midpoint of the interval. Changes rather than levels are reported because the coefficient in equation (1) of the comment is identified from differential movement across markets and not from permanent differences between them. Panel A restricts attention to spans of comparable length; Panel B reports every adjacent pair of waves, where the shorter spans in the extension period mechanically carry more sampling noise. The correlations over the decades that identify the published result run between 0.12 and 0.28, so the measure was never a strong predictor of changes even in the original period.

OA.2.5 Splitting the post period

The later window spans the pandemic, and a reader may reasonably ask whether the pandemic marks the break. It does not. For 2015 to 2019 the marriage coefficient is −0.0259 with a standard error of 0.0202, and for 2022 to 2024 it is −0.0016 with a standard error of 0.0147. Neither is significant, and the two are within a standard error of one another. Under the argument of Section IV of the comment neither estimate is interpretable in any case, and we report the split only to close the question.

OA.2.6 Rebasing the shares to 1980

To test whether a fresher weighting scheme restores the measure, we rebuild the shift-share series on employment shares from the 1980 Census five percent state sample. We use the *OCC1990* and *IND1950* mappings for industry and occupation, and we build the 1980 share matrix across 17 industry categories. The builder reproduces the structure of the 1970 share matrix closely, with a mean absolute difference of 1.4×10^{-3}. Moving the base year to 1980 helps over 1980 to 2010. The within-market correlation rises from 0.740 to 0.809, and the check rises from 0.593 to 0.799. It does nothing in the later waves, where the within-market correlation is -0.007.

One result of the rebasing bears on the original study rather than on the extension. Estimated over 1980 to 2010 with contemporaneous 1980 shares, the marriage coefficient is -0.0623 with a standard error of 0.0091. The published result therefore survives on 1980 shares. It is not an artifact of the base year. That the 1980 weights do nothing after 2010 is what we would expect if the plateau in relative earnings, rather than the age of the weights, is what ended the identifying variation.

OA.2.7 A direct test of the change in the coefficient

We test the change in the coefficient across periods directly, rather than comparing two separately estimated numbers, by pooling the two windows and interacting the relative wage with an indicator for the later waves. The pooled model is equivalent to the separate period regressions and allows the errors to be clustered by state throughout. Table OA.5 reports the results for the primary outcomes, with and without the controls measured at the market level. For marriage, the interaction is $+0.0453$ with a standard error of 0.0137 and a p value of 0.002 with controls, and $+0.0382$ with a standard error of 0.0132 and a p value of 0.006 without them. Section IV of the comment explains what this test measures. It is a change in what the constructed regressor predicts and not a change in how women behave.

OA.3 Robustness: Controls Measured at the Market Level

Table OA.6 reports the sensitivity of the estimates to the controls measured at the market level, across all outcomes and estimation windows. The controls are the mean years of schooling of women (*yrsedfem*) and of men (*yrsedmale*) and the sex ratio in the market (*sexratio*), all built directly from raw IPUMS microdata. Adding them leaves every sign and every significance level unchanged in every window. Over 1980 to 2010 the marriage coefficient moves from -0.0428 without controls to -0.0502 with them. Over 2015 to 2024 it is small and insignificant either way,

Table OA.5: Test of the Change in the Relative Wage Coefficient after 2010

	Without controls		With controls	
	Log rel. wage	Post × log rel. wage	Log rel. wage	Post × log rel. wage
Married	−0.0428***	+0.0382***	−0.0502***	+0.0453***
	(0.0074)	(0.0132)	(0.0086)	(0.0137)
Divorced	+0.0231***	−0.0306***	+0.0211***	−0.0287***
	(0.0049)	(0.0072)	(0.0053)	(0.0074)
Never married	+0.0183***	−0.0067	+0.0304***	−0.0182*
	(0.0065)	(0.0098)	(0.0074)	(0.0109)
p value on the interaction				
Married	0.006		0.002	
Divorced	< 0.001		< 0.001	
Never married	0.500		0.100	
Observations	52,188		52,184	

Each row reports a single regression pooling the 1980 to 2010 and 2015 to 2024 waves. *Post* is an indicator for the 2015 to 2024 waves. The coefficient on the log relative wage gives the relationship over the original period and the coefficient on the interaction gives the change in that relationship after 2010. The relative wage, the controls, the race by state linear trends, and the education×race×state fixed effects are all interacted with *Post*, so the model is algebraically equivalent to estimating the two windows separately while allowing a state-clustered test of the difference. All specifications absorb education×race×state, education×year, race×year, state×year, and birthyr×year fixed effects plus race by state linear trends. Cells are weighted by female population and standard errors in parentheses are clustered by state. The columns headed *With controls* add mean years of schooling for women and men and the sex ratio, matching the specification in Table 1 of the comment. The columns headed *Without controls* omit them, which is the specification underlying Table 3 of the comment. Coefficients use the convention of the original study and are scaled to a 10 percent increase in the relative wage. Both columns reproduce the period coefficients of the corresponding main-text exhibit: the without-controls column returns the estimates underlying Table 3, and the with-controls column returns those in Table 1 (−0.0502 and −0.0049 for the married outcome, on 52,184 cells). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

at -0.0045 without controls and -0.0049 with them.

Table OA.6: Robustness: Extended Table 2 With and Without Market-Level Controls

	1980–2010		1980–2024		2015–2024	
	No ctrl	With ctrl	No ctrl	With ctrl	No ctrl	With ctrl
Married						
Rel. wage	-0.0428***	-0.0502***	-0.0277***	-0.0224***	-0.0045	-0.0049
	(0.0074)	(0.0086)	(0.0049)	(0.0056)	(0.0099)	(0.0095)
Divorced						
Rel. wage	0.0231***	0.0211***	-0.0023	-0.0008	-0.0075*	-0.0076*
	(0.0049)	(0.0053)	(0.0028)	(0.0029)	(0.0041)	(0.0041)
Never Married						
Rel. wage	0.0183***	0.0304***	0.0284***	0.0235***	0.0116	0.0122
	(0.0065)	(0.0074)	(0.0056)	(0.0051)	(0.0080)	(0.0077)
Observations	23,775	23,774	52,188	52,184	28,413	28,410

Coefficients use the paper's 10 percent convention. Standard errors (parentheses) clustered at the state level. *With ctrl* columns add yrsedfem, yrsedmale, and sexratio computed for women and men directly from raw IPUMS USA in every wave, with years of schooling mapped from detailed EDUCD codes. All columns absorb educ×race×state, educ×year, race×year, state×year, and birthyr×year fixed effects plus race-by-state linear trends. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

OA.4 Secondary Outcomes

This section extends the secondary outcomes of Shenhav (2021) through 2024: cohabitation and living arrangements (Table OA.7), the education of the spouse (Table OA.8), and female labor supply (Table OA.9). All regressions use the baseline fixed effects, the cell weighting, and clustering by state. The argument of Section IV of the comment applies to these outcomes without modification. The estimates we report for the later waves therefore describe the decay of the regressor and not a change in behavior.

OA.4.1 Cohabitation and Living Arrangements

Table OA.7 extends the cohabitation and living arrangement outcomes through 2024. For living with a husband present, the coefficient over 1980 to 2010 is -0.0188 and is insignificant, and it moves to $+0.0028$ in the later waves. Living alone gives $+0.0289$ over 1980 to 2010, $+0.0213$ over the full panel, and $+0.0186$ over 2015 to 2024, the last significant at $p < 0.05$. Official cohabitation is insignificant in every window. The population weighted means for the living arrangements differ slightly from the published ones because household definitions differ across the raw extracts, and the relative trends across windows are preserved.

OA.4.2 The Education of the Spouse

Table OA.8 extends the spouse education outcomes through 2024, among married women with an identified spouse. For the probability of having a more educated spouse the coefficients are $+0.0511$ published over 1980 to 2010, $+0.0265$ in our reproduction of the same period, $+0.0392$ over the full 1980 to 2024 panel, and $+0.0264$ over 2015 to 2024, the last significant at $p < 0.05$. The continuous gap between the husband's and the wife's education attenuates from $+0.2581$ published over 1980 to 2010 to $+0.0240$ in the later waves, where it is insignificant. The setting itself moved a great deal over the same interval. Average spouse education differences shifted from parity, at $+0.007$ years over 1980 to 2010, to wives holding 0.472 more years of schooling than husbands after 2010.

OA.4.3 Labor Supply

Table OA.9 extends the female labor supply outcomes through 2024: usual weekly hours, annual weeks worked, employment, and log earnings. Over 1980 to 2010 weekly hours among workers give $+0.988$ with a standard error of 0.208, against a published $+1.021$, and annual weeks worked give -0.280 against a published -0.274. Over 2015 to 2024 all six labor supply coefficients attenuate to insignificance. Our weeks worked series conditions on employed workers and excludes allocated records (*QWKSWORK1*), and it matches the published population means to within 0.5 percent.

OA.5 Supplementary Exhibits

This section collects the exhibits that trace the estimated coefficient across single survey waves and across expanding panel windows. The cross-sectional regressions within a single wave in Table OA.10 give positive coefficients on the relative wage over 1980 to 2010, which we read as overall prosperity across markets, and they drift toward zero after 2010. The expanding panel estimates in Figure OA.2 remain negative and significant through 2017 and attenuate to zero by 2024. Table OA.12 reports tests for a structural break, and Table OA.11 reports the heterogeneity by race and education. Both describe the same attenuation, which Section OA.2 attributes to the decay of the measure. Auxiliary Kitagawa (Kitagawa, 1955) decompositions of the coefficient and Oaxaca-Blinder (Oaxaca, 1973; Blinder, 1973; Neumark, 1988) decompositions of the demographic levels are archived in the replication package.

Table OA.7: Extended Impact of the Relative Wage on Cohabitation and Living Arrangements, 1980–2024 (extending Shenhav, 2021, Table 3)

	Author 1980–2010 (published)	Ours 1980–2010 (replication)	Ours 1980–2024 (extended)	Ours 2015–2024 (new only)
Cohabitation				
Official Rept.	−0.0045	−0.0024	−0.0051	−0.0100
	(0.0175)	(0.0121)	(0.0046)	(0.0073)
Observations	16,925	17,092	44,537	27,444
Only Other Adult in HH is:				
Husband	−0.0404***	−0.0188	−0.0333***	+0.0028
	(0.0109)	(0.0135)	(0.0087)	(0.0085)
Observations	23,573	23,940	52,762	28,822
Single Male	+0.0087**	−0.0011	+0.0002	−0.0030
	(0.0037)	(0.0030)	(0.0025)	(0.0049)
Observations	23,573	23,940	52,762	28,822
Female	+0.0122***	+0.0139***	+0.0077***	−0.0035
	(0.0030)	(0.0043)	(0.0026)	(0.0040)
Observations	23,573	23,940	52,762	28,822
Lives Alone				
Live Alone	+0.0138	+0.0289***	+0.0213***	+0.0186**
	(0.0131)	(0.0106)	(0.0063)	(0.0072)
Observations	23,573	23,940	52,762	28,822
Three or More Adults				
2+ Other Adults	+0.0058	−0.0219**	+0.0047	−0.0133
	(0.0115)	(0.0107)	(0.0045)	(0.0087)
Observations	24,036	23,940	52,762	28,822
Mean Y: Official Rept.	0.159	0.105	0.122	0.133
Mean Y: Husband	0.457	0.543	0.502	0.470
Mean Y: Single Male	0.066	0.033	0.033	0.033
Mean Y: Female	0.059	0.056	0.067	0.075
Mean Y: Live Alone	0.259	0.153	0.159	0.163
Mean Y: 2+ Other Adults	0.159	0.219	0.247	0.269

Each row is a separate weighted least squares regression of the indicated outcome on the relative potential wage, extending Shenhav (2021). Cells are weighted by the female population in the marriage market; standard errors (parentheses) clustered at the state level. All specifications absorb education×race×state, education×year, race×year, state×year, and birth cohort×year fixed effects plus race-by-state linear trends. The **Author** column uses familyoutcomes.dta with the average potential wage and controls for mean schooling and the sex ratio (published spec.). The **Ours** columns use new IPUMS USA microdata (1980–2024), the average potential wage, and raw-IPUMS schooling and sex-ratio controls. Coefficients represent the effect of a 10% increase in the relative potential wage. Mean Y is the population-weighted average of the outcome (actual share). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The official cohabitation report is available from 1990 only and is defined only for female household heads. These living-arrangement variables are *alternative measures* of the source concepts: while the coding errors of earlier drafts are corrected, our population-weighted levels still differ materially from the author's (compare the Author and Ours Mean Y rows), most likely because the released code does not specify the author's "adult" definition. The coefficients should be interpreted through their direction and evolution across windows rather than as an exact level reproduction.

Table OA.8: Extended Impact of the Relative Wage on Spousal Education, 1980–2024 (extending Shenhav, 2021, Table 4)

	Author 1980–2010 (published)	Ours 1980–2010 (replication)	Ours 1980–2024 (extended)	Ours 2015–2024 (new only)
Spouse Ed., Relative to Own				
Less	−0.0232*	−0.0263**	−0.0272***	−0.0086
	(0.0120)	(0.0130)	(0.0080)	(0.0087)
Observations	22,663	23,383	50,606	27,223
Same	−0.0279*	−0.0001	−0.0120	−0.0178
	(0.0166)	(0.0132)	(0.0074)	(0.0118)
Observations	22,663	23,383	50,606	27,223
More	+0.0511***	+0.0265**	+0.0392***	+0.0264**
	(0.0133)	(0.0128)	(0.0067)	(0.0104)
Observations	22,663	23,383	50,606	27,223
Spouse Minus Own Ed.				
Difference	+0.2581***	+0.1894***	+0.1529***	+0.0240
	(0.0633)	(0.0629)	(0.0467)	(0.0596)
Observations	22,663	23,383	50,606	27,223
Mean Y: Less	0.319	0.322	0.364	0.398
Mean Y: Same	0.357	0.358	0.357	0.357
Mean Y: More	0.324	0.320	0.278	0.245
Mean Y: Difference	0.017	0.007	-0.261	-0.472

Each row is a separate weighted least squares regression of the indicated outcome on the relative potential wage, extending Shenhav (2021). Cells are weighted by the female population in the marriage market; standard errors (parentheses) clustered at the state level. All specifications absorb education×race×state, education×year, race×year, state×year, and birth cohort×year fixed effects plus race-by-state linear trends. The **Author** column uses familyoutcomes.dta with the average potential wage and controls for mean schooling and the sex ratio (published spec.). The **Ours** columns use new IPUMS USA microdata (1980–2024), the average potential wage, and raw-IPUMS schooling and sex-ratio controls. Coefficients represent the effect of a 10% increase in the relative potential wage. Mean Y is the population-weighted average of the outcome (actual share). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Table 4 outcomes are defined over married women (identified spouse present) only. *Difference* = spouse years of schooling minus own.

Table OA.9: Extended Table 5: Labor Supply Outcomes and the Relative Potential Wage, 1980–2024

	Author	This paper		
	1980–2010	1980–2010	1980–2024	2015–2024
Conditional on Working				
Avg. hours/week				
Rel. wage	1.021***	0.9880***	0.1143	0.2453
	(0.224)	(0.2078)	(0.1756)	(0.2040)
Observations	—	23,412	51,107	27,695
Avg. weeks worked				
Rel. wage	−0.274	-0.2796	-0.7783**	0.3026
	(0.441)	(0.4239)	(0.3364)	(0.1852)
Observations	—	23,412	51,107	27,695
Avg. ln(weekly income)				
Rel. wage	0.059**	0.0127	-0.0306*	0.0237
	(0.023)	(0.0204)	(0.0180)	(0.0161)
Observations	—	23,374	50,951	27,577
Avg. ln(annual income)				
Rel. wage	0.054*	0.0024	-0.0597*	0.0335
	(0.030)	(0.0305)	(0.0305)	(0.0211)
Observations	—	23,374	50,951	27,577
Unconditional				
Share positive income				
Rel. wage	−0.011	-0.0433***	-0.0346***	0.0040
	(0.015)	(0.0139)	(0.0107)	(0.0083)
Observations	—	23,766	52,162	28,396
Share in labor force				
Rel. wage	0.001	-0.0210*	-0.0293***	0.0055
	(0.013)	(0.0125)	(0.0100)	(0.0066)
Observations	—	23,766	52,162	28,396

Author column: Shenhav (2021), Table 5. Standard errors (parentheses) clustered at the state level. All specifications include educ×race×state, educ×year, race×year, state×year, and birthyr×year fixed effects plus race-by-state linear trends. Outcomes are built from raw IPUMS USA data (1980–2024). Reported coefficients use the paper's convention for the effect of an approximately 10 percent increase in the relative potential wage. Allocated weeks-worked observations are excluded using QWKSWORK1. Weeks worked is conditional on workers in both columns.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA.10: Wave-specific cross-sectional relationship between the relative wage and marriage outcomes, 1980–2024.

Wave	Share married $\hat{\beta}_t$	SE	Share divorced $\hat{\beta}_t$	SE	Share never married $\hat{\beta}_t$	SE
1980	$+0.0194^{***}$	(0.0043)	$+0.0069^{***}$	(0.0023)	-0.0225^{***}	(0.0032)
1990	$+0.0274^{***}$	(0.0052)	$+0.0076^{***}$	(0.0027)	-0.0343^{***}	(0.0046)
2000	$+0.0313^{***}$	(0.0065)	$+0.0051$	(0.0036)	-0.0361^{***}	(0.0055)
2010	$+0.0185^{***}$	(0.0055)	$+0.0071^{**}$	(0.0035)	-0.0257^{***}	(0.0052)
2015	$+0.0154^{**}$	(0.0060)	$+0.0048$	(0.0037)	-0.0191^{***}	(0.0052)
2017	$+0.0161^{***}$	(0.0051)	$+0.0024$	(0.0034)	-0.0193^{***}	(0.0050)
2019	$+0.0053$	(0.0045)	$+0.0065^{*}$	(0.0032)	-0.0122^{***}	(0.0044)
2022	$+0.0026$	(0.0054)	$+0.0074^{***}$	(0.0023)	-0.0093^{*}	(0.0049)
2024	-0.0044	(0.0043)	$+0.0048^{**}$	(0.0019)	-0.0016	(0.0039)

Coefficients from separate OLS regressions within each ACS/Census wave. Each $\hat{\beta}_t$ is the cross-market slope of the outcome on the log relative potential wage, rescaled to represent the effect of a 10 percent increase. Fixed effects: education, race, state, birth cohort (separable). Weights: female cell population. Standard errors clustered at the state level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA.11: Heterogeneity of the wage–marriage relationship by education–race group, pre and post 2010.

Group	Share married Pre 1980–2010	Share married Post 2015–2024	Share divorced Pre 1980–2010	Share divorced Post 2015–2024	Share never married Pre 1980–2010	Share never married Post 2015–2024
HS or less / White	-0.0422^{**}	-0.0075	-0.0079	-0.0085	$+0.0489^{***}$	$+0.0146$
HS or less / Black	-0.0874^{***}	-0.0075	$+0.0082$	-0.0032	$+0.0820^{**}$	-0.0073
HS or less / Hispanic	$+0.0119$	$+0.0009$	$+0.0051$	-0.0027	-0.0152	-0.0002
Some college / White	-0.0501^{*}	-0.0041	$+0.0310$	-0.0033	$+0.0167$	$+0.0072$
Some college / Black	-0.0131	$+0.0050$	-0.0265	-0.0027	$+0.0284$	-0.0070
Some college / Hispanic	$+0.0123$	-0.0130	$+0.0066$	$+0.0013$	-0.0164^{**}	$+0.0115$

Each entry is the OLS coefficient on the log relative potential wage from a regression run separately within each education-race group and time period. Fixed effects: state×year, birth-cohort×year, state. Weights: female cell population. Standard errors clustered at the state level. All coefficients rescaled to the effect of a 10 percent increase in the relative wage. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table OA.12: Structural-break test: did the wage–marriage relationship change after 2010?

	Share married Coeff.	SE	Share divorced Coeff.	SE	Share never married Coeff.	SE
$\hat{\beta}$ (pre-2015, base)	-0.0428^{***}	(0.0074)	$+0.0231^{***}$	(0.0049)	$+0.0183^{***}$	(0.0065)
$\hat{\delta}$ (post-2015 change)	$+0.0382^{***}$	(0.0132)	-0.0306^{***}	(0.0072)	-0.0067	(0.0098)
$\hat{\beta}+\hat{\delta}$ (post-2015)	-0.0045	(0.0099)	-0.0075	(0.0041)	$+0.0116$	(0.0080)
N	52,188		52,188		52,188	

Pooled regression on the full 1980–2024 panel. $\hat{\beta}$ is the pre-2015 slope of the outcome on the log relative potential wage. $\hat{\delta}$ is the additional slope for waves 2015–2024 (estimated from the interaction wage × 1[year≥2015]). A significant $\hat{\delta}$ indicates a structural break in the wage–marriage relationship after 2010. All coefficients rescaled to the effect of a 10 percent increase in the relative wage. Fixed effects: education×race×state, education×year, race×year, state×year, birth-cohort×year. Education×race×state effects and race×state linear trends are fully interacted with era. Weights: female cell population. Standard errors clustered at the state level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

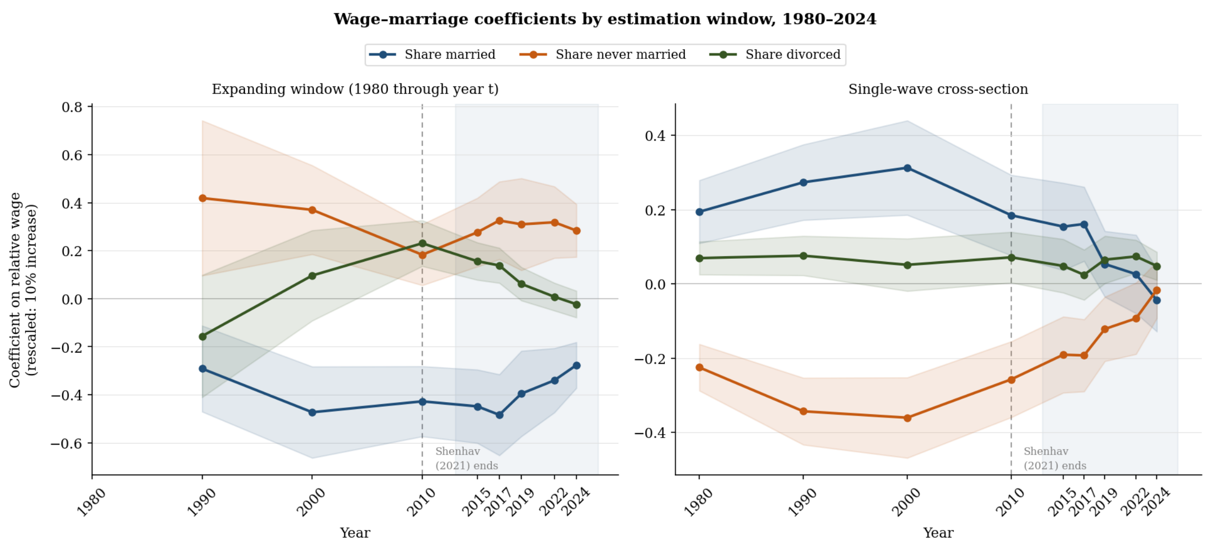

Figure OA.2: Coefficients relating the wage to marriage by estimation window, 1980 to 2024. Each series shows the coefficient by ordinary least squares on the female-to-male relative potential wage, rescaled to a 10 percent increase, for the three marriage outcomes. The left panel uses an expanding window, including all waves from 1980 through year t, with the full set of fixed effects and race by state trends. The right panel reports cross-sectional estimates within a single wave with separable fixed effects. Shaded bands are 95 percent confidence intervals and standard errors are clustered by state. The vertical dashed line marks 2010 and the shaded band is the extension period.

OA.6 Contents of the Replication Package

Complete reproduction files for the baseline exhibits in Shenhav (2021) are archived in the online replication package rather than reprinted here. The package reproduces all six main tables, 15 appendix tables, and six figures in Shenhav (2021). Every specification estimates equation (1) on the author's microdata files and matches the published point estimates to three decimal places. The package also includes the extraction scripts for IPUMS USA and IPUMS CPS, the 1980 Census five percent extract, the occupation crosswalks, the top coding routines, the scripts that build the base year share matrices, and the Stata and Python code that reproduces every table and figure.

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